

Sharp jumps in z_{it} : countries, dates, and causes

2026-09-11 panel, RE intercepts specification

1 Setup

The fitted trend is linear in calendar time (common g , country intercept a_i). A cliff in the plotted cycle is therefore a cliff in log real GDP per worker. Consecutive-quarter changes with $|\Delta y| \geq 0.25$ (about 28% QoQ) were flagged and split into local GDP, USD GDP, employment, and the base-year exchange rate.

2 Main finding

Currency units are not the main story. The base-year exchange rate is a country constant; its log-difference is zero at every flagged jump. Known redenominations (Romania 2005, Belarus 2016, and others) are smooth in this file. The cliffs on the cycle plot are almost entirely **employment scale or source breaks**, plus one **GDP units break** (El Salvador, 2005).

3 Distinct level shifts

These are the one-time (or paired) cliffs, not repeating seasonality. $\Delta \log y$ is the change in log real GDP per worker from the previous quarter.

Country	Date	$\Delta \log y$	Driver	What moved	Cause
El Salvador	2004Q4→2005Q1	+3.95 ($\times 52$)	GDP	GDP 85 → 4352; empl flat; NER = 1	IFS real GDP units/scale break at the 2005 base year (already dollarized; not a FX reform)
Slovak Rep.	1999Q4→2000Q1	+1.38	employment	empl 2118 → 524 ($\div 4$)	employment $4\times$ too small for 2000–01

Country	Date	$\Delta \log y$	Driver	What moved	Cause
Slovak Rep.	2001Q4→2002Q1	−1.45	employment	empl 535 → 2105 ($\times 4$)	scale reverts
Israel	2016Q4→2017Q1	+1.36	employment	empl 3756 → 944 ($\div 4$)	same $4\times$ break for 2017 only
Israel	2017Q4→2018Q1	−1.41	employment	empl 959 → 3865 ($\times 4$)	scale reverts
Montenegro	2022Q4→2023Q1	−1.81	employment	empl 256 → 1113 ($\times 4.3$)	2023Q1–Q4 stuck at 1113.145 (about $4\times$ too high)
Montenegro	2023Q4→2024Q1	+1.03	employment	empl 1113 → 281 ($\div 4$)	scale reverts
Peru	2010Q4→2011Q1	−1.31	employment	empl 4424 → 16136 ($\times 3.65$)	employment definition/coverage jump (not a clean $4\times$)
Seychelles	2015Q4→2016Q1	+1.03	employment	empl 27.9 → 10.4	annual interpolation → quarterly employment
Seychelles	2016Q4→2017Q1	−1.24	employment	empl 11.3 → 39.3 ($\times 3.5$)	employment jumps back up; GDP flat
Paraguay	2009Q4→2010Q1	+1.12	employment	empl 2850 → 925	annual interpolation → quarterly employment
Paraguay	2016Q4→2017Q1	−0.88	employment	empl 1100 → 2732 ($\times 2.5$)	employment scale/source switch
Bolivia	1996Q4→1997Q1	−0.52	employment	empl 1729 → 2758; GDP flat	employment level break
Dominican Rep.	2003–2015, many quarters	± 0.67 to ± 0.76	employment	quarterly $\sim 1.7\text{m}$ vs annual-interp $\sim 3.5\text{m}$	repeating $\sim 2\times$ sawtooth when the employment source flips; not a one-off jump
Macao	2019Q4→2020Q1	−0.66	GDP	GDP halves; empl flat	COVID tourism collapse (real)
Egypt	2022Q2→2022Q3	+0.58	GDP	GDP $1.16 \times 10^6 \rightarrow 2.09 \times 10^6$; empl flat	one-quarter GDP level jump; not employment and not NER
Romania	Q4→Q1 in 1997, 1998, 1999, 2009	about −0.5	GDP	GDP, not employment	NSA seasonal drop (and 2009); not the 2005 leu reform

4 Notes

Slovak Republic and Israel are the same $4\times$ employment-unit bug already patched in the SOE extract; it is back in this file. Montenegro 2023 looks like a single annual number copied into every quarter.

NSA seasonality (Romania, Bulgaria, Montenegro, Mongolia, and others, with $|\Delta y| \approx 0.25\text{--}0.35$ every winter) is a repeating pattern, not a one-time cliff.

Full consecutive-quarter list: z-jumps.csv (211 rows with $|\Delta y| \geq 0.25$). Distinct events: jump-events.csv.